ESP32 Autonomous Smart Grid Controller

Comprehensive Hardware Architecture, Circuit Interfaces & Line-by-Line Firmware Analysis

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Target File: firmware/.../smartgrid_closed_loop_esp32.ino
Hardware Platform: ESP32-WROOM-32 (Xtensa Dual-Core 240 MHz)
Firmware Version: 2.0.0-closed-loop

1. System Hardware Architecture & Functional Design

The microgrid embedded firmware operates as a real-time, closed-loop cyber-physical controller implemented on an Espressif ESP32 dual-core microcontroller clocked at 240 MHz. The system coordinates high-precision high-side DC current and voltage sensing across five discrete microgrid feeders, drives a 4-channel optoisolated power switchgear relay module, manages a 128x64 SPI OLED telemetry display, communicates via secure TLS MQTT (port 8883) with a cloud digital twin SCADA layer, and enforces an autonomous sub-second hardware breaker trip directly at the edge.

Subsystem	Components & Protocols	Key GPIO Pinout	Operational Role
Main MCU	ESP32-WROOM-32 (Dual Core 240MHz)	Onboard 3.3V Logic	Sensing, safety logic, TLS telemetry, actuator control
I2C Bus 1 (Wire)	4x Texas Instruments INA219	SDA: GPIO 27 SCL: GPIO 33	Monitors House (0x40), Industry (0x41), Street (0x44), Hospital (0x45)
I2C Bus 2 (I2C2)	1x Texas Instruments INA219	SDA: GPIO 25 SCL: GPIO 32	Monitors Critical Infrastructure (0x40) avoiding address clash
Relay Switchgear	4-Ch Optoisolated Relays (Active-LOW)	Hosp: 14, Strt: 13 Hous: 4, Indu: 26	Feeder disconnection/reconnection; fail-safe normally energized
Source Sensing	Digital Rail Presence Sense	SMPS1: 34, SMPS2: 35 Battery: 39	Detects Dual SMPS (5A), Single SMPS (1A), or Battery (800mA)
Display (SPI)	0.96 inch SSD1306 OLED (128x64)	MOSI: 23, CLK: 18, DC: 21 CS: 19, RESET: 22	Real-time telemetry, relay vector [1111], mode, and alerts
Cloud Networking	WiFiClientSecure + PubSubClient	802.11 b/g/n (2.4 GHz)	HiveMQ Cloud TLS MQTTS (Port 8883), 1Hz pub, sub control

2. Line-by-Line Firmware Dissection & Electrical Rationale

Part 1: Header Inclusions & External Libraries (Lines 1–12)

```
1: #include <Arduino.h>
2: #include <Wire.h>
3: #include <SPI.h>
4: #include <WiFi.h>
5: #include <WiFiClientSecure.h>
6: #include <PubSubClient.h>
7: #include <ArduinoJson.h>
8: #include <Adafruit_GFX.h>
9: #include <Adafruit_SSD1306.h>
10: #include <Adafruit_INA219.h>
11: #include <time.h>
```

Line 1 (<Arduino.h>): Core framework providing hardware register bindings, pin mapping constants, and core timing routines (millis(), digitalRead(), digitalWrite()).

Line 2 (<Wire.h>): Hardware I2C communications engine used to read shunt voltage, bus voltage, and current from INA219 sensor ICs.

Line 3 (<SPI.h>): Serial Peripheral Interface master driver enabling high-throughput block data streaming to the OLED display.

Lines 4–5 (<WiFi.h>, <WiFiClientSecure.h>): ESP32 802.11 station stack plus mbedTLS cryptographic socket wrapper supporting TLS 1.2/1.3 mutual handshake on port 8883.

Line 6 (<PubSubClient.h>): Lightweight MQTT client handling binary packet encoding, keep-alives, QoS handshakes, and subscription callbacks.

Line 7 (<ArduinoJson.h>): Fixed-memory JSON serialization/deserialization library avoiding heap fragmentation during telemetry packaging.

Lines 8–10 (<Adafruit_GFX.h>, <Adafruit_SSD1306.h>, <Adafruit_INA219.h>): Peripheral device drivers for the SSD1306 display controller and INA219 current monitors.

Line 11 (<time.h>): POSIX standard time library interfacing with ESP32 SNTP client for sub-millisecond clock synchronization.

Part 2: SPI OLED Display Configuration (Lines 13–26)

```
16: #define SCREEN_WIDTH 128
17: #define SCREEN_HEIGHT 64
19: #define OLED_MOSI  23
20: #define OLED_CLK   18
21: #define OLED_DC    21
22: #define OLED_CS    19
23: #define OLED_RESET 22
25: Adafruit_SSD1306 display(SCREEN_WIDTH, SCREEN_HEIGHT, OLED_MOSI, OLED_CLK, OLED_DC, OLED_RESET, OLED_CS);
```

Lines 16–17: Sets display dimensions to 128x64 monochrome pixels (requiring a 1024-byte in-memory frame buffer).

Lines 19–20: Maps SPI data lines: MOSI to GPIO 23 and Clock (SCK) to GPIO 18.

Line 21: Maps Data/Command select pin to GPIO 21 (High = Pixel framebuffer stream, Low = SSD1306 command byte).

Lines 22–23: Maps Chip Select (CS) to GPIO 19 and Hardware Reset to GPIO 22.

Line 25: Instantiates the global display driver object. Using hardware-mapped SPI pins guarantees high frame refresh rates without blocking core control tasks.

Part 3: Relay Switchgear & Source Sense Pin Definitions (Lines 27–42)

```
30: #define RELAY_HOUSE      4
31: #define RELAY_INDUSTRY   26
32: #define RELAY_STREET     13
33: #define RELAY_HOSPITAL   14
35: #define RELAY_ON  LOW
36: #define RELAY_OFF HIGH
39: #define SMPS1_PIN        34
40: #define SMPS2_PIN        35
41: #define BATTERY_PIN      39
```

Lines 30–33: Assigns relay coil control lines to GPIO 4 (House), GPIO 26 (Industry), GPIO 13 (Street Light), and GPIO 14 (Hospital Auxiliary).

Lines 35–36 (Active-LOW Logic): Standard optocoupler boards are active-LOW: pulling the GPIO to 0V (LOW) drives optodiode forward current, energizing the relay coil. Setting the GPIO to 3.3V (HIGH) turns off the optocoupler, de-energizing the contacts.

Lines 39–41: Configures digital sense lines: GPIO 34 (SMPS 1), GPIO 35 (SMPS 2), and GPIO 39 (Battery). These are input-only pins (GPI) on the ESP32 with no internal pull-ups, requiring external pull-down resistors.

Part 4: Dual I2C Bus Architecture & INA219 Addressing (Lines 43–53)

```
46: TwoWire I2C2 = TwoWire(1);
48: Adafruit_INA219 inaHouse(0x40);     // On Wire (SDA: 27, SCL: 33)
49: Adafruit_INA219 inaIndustry(0x41);  // On Wire (SDA: 27, SCL: 33)
50: Adafruit_INA219 inaStreet(0x44);    // On Wire (SDA: 27, SCL: 33)
51: Adafruit_INA219 inaHospital(0x45);  // On Wire (SDA: 27, SCL: 33)
52: Adafruit_INA219 inaCritical(0x40);  // On I2C2 (SDA: 25, SCL: 32)
```

Line 46 (Dual I2C Bus Instantiation): The ESP32 contains two hardware I2C peripherals. TwoWire(1) instantiates the secondary controller I2C2.

Lines 48–51: Mounts four INA219 sensors on primary bus Wire using distinct address pin configurations: 0x40 (GND/GND), 0x41 (GND/VCC), 0x44 (VCC/GND), 0x45 (VCC/VCC).

Line 52 (Critical Address Conflict Resolution): The fifth sensor (inaCritical) shares address 0x40. Placing it on I2C2 (SDA: 25, SCL: 32) eliminates bus contention without requiring an external multiplexer IC (like TCA9548A).

Part 5: Secure Cloud Network & HiveMQ MQTT Credentials (Lines 54–69)

```
57: const char* ssid              = "Verizon-SM-N981U-7692";
58: const char* password          = "nzjf807(";
60: const char* mqtt_server       = "78d11d43ac27466695cd0e4f5e7405c8.s1.eu.hivemq.cloud";
61: const int   mqtt_port         = 8883;
62: const char* mqtt_user         = "saptarshidas";
63: const char* mqtt_password     = "saptarshi2007";
64: const char* mqtt_topic        = "smartgrid/live";
65: const char* mqtt_control_topic = "smartgrid/control";
67: WiFiClientSecure espClient;
68: PubSubClient client(espClient);
```

Lines 57–58: Wi-Fi Station configuration parameters.

Lines 60–63: HiveMQ Cloud TLS cluster endpoint, port 8883 (MQTTS), and secure user authentication credentials.

Lines 64–65: Dedicated MQTT pub/sub topics: smartgrid/live (1Hz telemetry uplink) and smartgrid/control (closed-loop command downlink).

Lines 67–68: Instantiates WiFiClientSecure and binds it to PubSubClient for encrypted payload transport.

Part 6: State Tracking, Timebase & NTP Synchronization (Lines 70–95)

```
73: bool relay_hospital = true;
74: bool relay_street   = true;
75: bool relay_house    = true;
76: bool relay_industry = true;
78: char lastActionMsg[32] = "READY (30s OBS)";
79: unsigned long lastSampleTime = 0;
80: const unsigned long SAMPLE_INTERVAL = 1000;
83: String getFormattedTimestamp() {
84:     struct tm timeinfo;
85:     if (getLocalTime(&timeinfo, 100)) {
86:         char buffer[32];
87:         strftime(buffer, sizeof(buffer), "%Y-%m-%d %H:%M:%S", &timeinfo);
88:         return String(buffer);
89:     }
90:     char fallback[32];
91:     unsigned long sec = millis() / 1000;
92:     snprintf(fallback, sizeof(fallback), "2026-09-13 %02lu:%02lu:%02lu", (sec / 3600) % 24, (sec / 60) % 60, sec % 60)
93:     return String(fallback);
94: }
```

Lines 73–76: In-memory state tracking variables. All initialized to true (energized at boot).

Line 78: Status string buffer for the OLED display. Initialized to 'READY (30s OBS)' for the initial observation window.

Lines 79–80: Non-blocking timer tracking variables enforcing a 1.0 Hz (1000 ms) cyclic timebase using millis().

Lines 83–89: Retrieves real-time clock from SNTP via getLocalTime() and formats as YYYY-MM-DD HH:MM:SS.

Lines 90–94 (Fault-Tolerant Fallback): If NTP sync is pending or network is down, generates a continuous synthetic timestamp from millis() uptime to prevent null values in downstream SCADA databases.

Part 7: Client ID Generation & Relay Hardware Driver (Lines 96–111)

```
97: String getUniqueClientID() {
98:     uint8_t mac[6];
99:     WiFi.macAddress(mac);
100:     char clientId[32];
101:     snprintf(clientId, sizeof(clientId), "SMARTGRID_%02X%02X%02X%02X%02X%02X", mac[0], mac[1], mac[2], mac[3], mac[4]
102:     return String(clientId);
103: }
105: void applyRelays() {
106:     digitalWrite(RELAY_HOSPITAL, relay_hospital ? RELAY_ON : RELAY_OFF);
107:     digitalWrite(RELAY_STREET,   relay_street   ? RELAY_ON : RELAY_OFF);
108:     digitalWrite(RELAY_HOUSE,    relay_house    ? RELAY_ON : RELAY_OFF);
109:     digitalWrite(RELAY_INDUSTRY, relay_industry ? RELAY_ON : RELAY_OFF);
110: }
```

Lines 97–103: Extracts the ESP32 factory MAC address and generates a unique MQTT Client ID (e.g., SMARTGRID_A4CF129B43E0), preventing broker session conflicts.

Lines 105–110 (Hardware Sync Driver): Synchronizes the four logical Boolean state flags with the physical GPIO output pins using active-LOW mapping.

Part 8: Closed-Loop Actuator Callback (Lines 112–150)

```
115: void mqttCallback(char* topic, byte* payload, unsigned int length) {
116:     JsonDocument doc;
117:     DeserializationError err = deserializeJson(doc, payload, length);
118:     if (err) return;
123:     const char* cmd    = doc["cmd"] | "";
124:     const char* load   = doc["load"] | "";
125:     int state          = doc["state"] | -1;
126:     const char* reason = doc["reason"] | "";
131:     if (state == 0 || state == 1) {
132:         bool targetState = (state == 1);
133:         uint8_t pinVal = targetState ? RELAY_ON : RELAY_OFF;
135:         if (strcasecmp(load, "hospital") == 0) {
136:             relay_hospital = targetState;
137:             digitalWrite(RELAY_HOSPITAL, pinVal);
138:         } else if (strcasecmp(load, "street") == 0) { ... }
148:         snprintf(lastActionMsg, sizeof(lastActionMsg), "%s %s", cmd, load);
149:     }
150: }
```

Line 115: Asynchronous interrupt handler triggered whenever an MQTT control packet arrives.

Lines 116–118: Parses incoming JSON bytes. If corrupted, exits immediately without hanging.

Lines 123–126: Safely extracts action parameters: cmd (SHED/RESTORE), load (target branch), state (0/1), and reason (diagnostic log).

Lines 131–147: Validates state, identifies target load via case-insensitive matching (strcasecmp), updates the state variable, and executes physical relay actuation.

Line 148: Updates OLED display with the executed action string.

Part 9: Wi-Fi & MQTT Auto-Reconnection Engine (Lines 151–170)

```
152: void connectWiFi() {
153:     WiFi.mode(WIFI_STA);
154:     WiFi.setAutoReconnect(true);
155:     WiFi.persistent(true);
156:     WiFi.begin(ssid, password);
157: }
159: void reconnectMQTT() {
160:     if (WiFi.status() != WL_CONNECTED) return;
161:     if (!client.connected()) {
162:         String clientId = getUniqueClientID();
163:         if (client.connect(clientId.c_str(), mqtt_user, mqtt_password)) {
165:             client.subscribe(mqtt_control_topic);
166:         }
167:     }
168: }
```

Lines 152–157: Sets Wi-Fi station mode, enables hardware auto-reconnect, and stores credentials in NVS flash.

Lines 159–168: Verifies physical Layer-3 connection before attempting broker handshake. Upon connection, re-subscribes to smartgrid/control to restore the closed-loop control link.

Part 10: Local Autonomous Protection & Edge Breaker (Lines 171–231)

```
174: const float THRESH_HOSPITAL = 272.0;
175: const float THRESH_STREET   = 405.0;
176: const float THRESH_HOUSE    = 256.0;
177: const float THRESH_INDUSTRY = 264.0;
178: const float THRESH_CRITICAL = 320.0;
180: void evaluateLocalAutonomousProtection(...) {
181:     bool cloudActive = (WiFi.status() == WL_CONNECTED && client.connected());
185:     if (c_hosp > THRESH_HOSPITAL && relay_hospital) {
186:         relay_hospital = false;
187:         digitalWrite(RELAY_HOSPITAL, RELAY_OFF);
188:         snprintf(lastActionMsg, sizeof(lastActionMsg), "LOCAL TRIP HOSP");
189:     }
...
213:     if (!cloudActive && limit_mA > 0 && tot_mA >= (0.85 * limit_mA)) {
214:         if (relay_industry) { relay_industry = false; digitalWrite(RELAY_INDUSTRY, RELAY_OFF); }
219:         else if (relay_house) { relay_house = false; digitalWrite(RELAY_HOUSE, RELAY_OFF); }
224:         else if (relay_street) { relay_street = false; digitalWrite(RELAY_STREET, RELAY_OFF); }
230:     }
231: }
```

Lines 174–178: Empirically calibrated 2-standard-deviation (2σ) overcurrent thresholds in mA.

Lines 185–208 (Sub-Second Edge Overcurrent Breaker): If any branch current breaches its 2σ limit, the relay is de-energized immediately (<1.0 s). Runs locally on the microcontroller regardless of Wi-Fi status.

Lines 213–230 (Autonomous Offline Capacity Shedding): If offline and aggregate load reaches $\ge 85\%$ of source capacity, the microcontroller sheds lower-priority loads locally: Industry (P1) $\to$ House (P2) $\to$ Street (P3). Hospital (P4) and Critical Infrastructure remain protected.

Part 11: Fault-Tolerant INA219 Shunt Driver (safeRead) (Lines 232–252)

```
233: void safeRead(TwoWire &bus, Adafruit_INA219 &sensor, uint8_t addr, float &voltage, float &current_mA, float &power_mW
234:     bus.beginTransmission(addr);
235:     if (bus.endTransmission() != 0) {
236:         ok = false; voltage = 0.0; current_mA = 0.0; power_mW = 0.0; return;
238:     }
241:     voltage = sensor.getBusVoltage_V();
242:     current_mA = sensor.getCurrent_mA();
244:     if (isnan(voltage) || isnan(current_mA)) {
245:         ok = false; voltage = 0.0; current_mA = 0.0; power_mW = 0.0;
247:     } else {
248:         ok = true;
249:         if (current_mA > -0.5 && current_mA < 0.5) current_mA = 0.0;
250:         power_mW = voltage * current_mA;
251:     }
252: }
```

Lines 234–239: Probes I2C device acknowledgment via endTransmission(). If NACKed (disconnected/shorted), zeroes variables and returns without locking the bus.

Lines 241–247: Reads voltage and current; validates against floating-point NaN values.

Line 249 (Zero-Drift Noise Filter): Clamps small thermal shunt offset noise ($\pm 0.5\text{ mA}$) to zero.

Line 250: Computes true instantaneous branch power ($P = V \times I$) in milliwatts (mW).

Part 12: OLED Graphical Status Dashboard (updateOLED) (Lines 253–289)

```
256: display.clearDisplay();
260: bool wifiOk = (WiFi.status() == WL_CONNECTED);
261: bool mqttOk = client.connected();
262: const char* netTag = (wifiOk && mqttOk) ? "CLOUD" : (wifiOk ? "NO-MQTT" : "OFFLINE");
264: display.printf("%s|%4.0f|%s", modeStr, limit_mA, netTag);
267: display.printf("HOS: %4.0fmA STR: %4.0fmA", c_hosp, c_str);
270: display.printf("HOU: %4.0fmA IND: %4.0fmA", c_hou, c_ind);
273: display.printf("CRI: %4.0fmA TOT:%5.0fmA", c_crit, tot_mA);
277: display.printf("RLY:[%d%d%d%d] (%s)", r_hosp, r_str, r_hou, r_ind, (tot_mA > 0.85*limit_mA) ? "SHED" : "NORM");
285: display.printf("ACT: %s", lastActionMsg);
287: display.display();
```

Lines 260–262: Resolves 3-tier network status: CLOUD, NO-MQTT, or OFFLINE.

Lines 264–285: Formats 6 rows of telemetry: source mode, branch currents, aggregate current, 4-bit relay state vector, load status, and recent actuation message.

Line 287: Flushes the 1024-byte framebuffer over SPI to the SSD1306 panel.

Part 13: System Initialization (setup) (Lines 290–357)

```
294: Serial.begin(115200);
298: pinMode(RELAY_HOSPITAL, OUTPUT); ... pinMode(RELAY_INDUSTRY, OUTPUT);
307: applyRelays(); // ALL 4 ON at boot
310: pinMode(SMPS1_PIN, INPUT); pinMode(SMPS2_PIN, INPUT); pinMode(BATTERY_PIN, INPUT);
315: display.begin(SSD1306_SWITCHCAPVCC);
328: Wire.begin(27, 33); Wire.setTimeOut(50);
330: I2C2.begin(25, 32); I2C2.setTimeOut(50);
333: inaHospital.begin(&Wire); ... inaCritical.begin(&I2C2);
340: espClient.setInsecure();
341: client.setServer(mqtt_server, mqtt_port); client.setCallback(mqttCallback); client.setBufferSize(1024);
345: connectWiFi();
347: configTime(19800, 0, "pool.ntp.org", "time.google.com");
```

Lines 298–307: Configures relay GPIOs as outputs and energizes all loads for the 30s boot observation phase.

Lines 328–331: Initializes both hardware I2C controllers with 50 ms timeouts to guard against bus freezes.

Lines 333–337: Initializes all five INA219 sensors across the two buses.

Lines 340–343: Sets TLS to accept server certificates, registers mqttCallback, and increases MQTT buffer to 1024 bytes.

Line 347: Initiates SNTP time synchronization (+5:30 IST = 19,800 seconds offset).

Parts 14 & 15: Main Execution Loop, Telemetry Publishing & Serialization (Lines 358–503)

```
362: if (WiFi.status() == WL_CONNECTED) {
363:     if (!client.connected()) reconnectMQTT();
364:     client.loop();
365: } else { ... non-blocking reconnect every 5s ... }
375: if (millis() - lastSampleTime >= SAMPLE_INTERVAL) {
382:     if (smps1 && smps2) { modeStr = "DUAL SMPS"; current_limit = 5000.0; }
387:     else if (smps1 || smps2) { modeStr = "SINGLE SMPS"; current_limit = 1000.0; }
390:     else if (battery) { modeStr = "BATTERY"; current_limit = 800.0; }
403:     safeRead(I2C2, inaCritical, ...); safeRead(Wire, inaHospital, ...); ...
409:     float total_mA = c_crit + c_hosp + c_str + c_hou + c_ind;
413:     evaluateLocalAutonomousProtection(...);
422:     if (client.connected()) {
423:         JsonDocument doc;
424:         doc["timestamp"] = getFormattedTimestamp(); ...
493:         client.publish(mqtt_topic, jsonBuffer, len);
494:     }
501:     updateOLED(...);
502: }
```

Lines 362–373: Maintains MQTT loop and handles non-blocking Wi-Fi auto-recovery.

Line 375: Enforces 1.0 Hz sampling using millis().

Lines 382–393: Dynamic capacity allocation: Dual SMPS (5000 mA), Single SMPS (1000 mA), or Battery (800 mA).

Lines 403–407: Reads telemetry from all five sensors using safeRead().

Line 413: Runs local edge protection to trip overcurrent in <1.0 s if thresholds are breached.

Lines 422–494: Serializes complete nested JSON payload and publishes to smartgrid/live.

Line 501: Refreshes OLED display with latest readings and relay status.

3. Summary of Electrical & Engineering Decisions

Challenge	Firmware Implementation	Engineering Rationale
I2C Address Collision	Dual Hardware Buses (Wire & I2C2)	inaCritical & inaHouse share address 0x40. Isolating inaCritical on I2C2 (GPIO 25/32) avoids collisions without extra hardware.
Relay Power-on Glitches	Active-LOW Logic (LOW = ON)	Matches optocoupler input biasing on standard relay boards, preventing spurious relay chatter during power-on boot states.
Edge Protection Latency	< 1.0s Local 2-Sigma Breaker	Runs locally on the microcontroller, guaranteeing hardware trip protection even if Wi-Fi drops or the host controller freezes.
Switchgear Chattering	15.0s Governor Dwell Time	Prevents rapid cycling during marginal capacity fluctuations, reducing mechanical contactor wear and contact welding.
Sensor Zero Drift	Clamping [-0.5, +0.5] mA to 0.0	Eliminates thermal drift noise on INA219 shunt readings when circuits are in a zero-current state.